POLICY RESEARCH / FOOD SYSTEM PRODUCTIVITY

THE MARGINAL-COST REVOLUTION IN PREPARED FOOD

A UK food manufacturing model for high-quality, affordable prepared meals

A costed menu, marginal-cost framework and regional production proposal for HM Treasury

Independent LSE-style analytical memorandum | 5 November 2026

Prepared for policy discussion. This paper is independently prepared in an economics-research format; it is not produced by, commissioned by, or endorsed by the London School of Economics and Political Science (LSE) or HM Treasury.

THE PROPOSITION

Four proposed 450 g meals can be modelled at £0.91-£1.61 in ingredients per portion after a 5% trim/waste allowance. Under the stated high-throughput assumptions, variable factory-delivered cost is £1.74-£2.45. A matched operating scenario suggests approximately 49p lower variable cost per meal - an opportunity to test, not a proven national saving.

Prepared-food affordability is not solved by cheaper ingredients alone. The opportunity is to combine recipe engineering, pooled procurement, predictable demand, process automation, low waste and regional cold-chain distribution while protecting taste, nutrition, jobs and supplier competition.

Scope: Great Britain, with UK-wide official statistics where available. Prices are public-listing benchmarks accessed around 5 November 2026; no supplier has quoted or endorsed the model.

Executive summary

This report tests whether high-quality prepared food can be sold at everyday prices by redesigning how meals are specified, purchased, cooked, packed and distributed. It builds four illustrative British-inspired recipes, costs them against publicly visible UK wholesale listings, and models a regional production hub capable of making approximately 10,000 meals a day.

The economic case is not that ingredients are suddenly cheap. It is that a prepared meal contains several avoidable or scale-sensitive costs beyond ingredients: labour minutes, trim and spoilage, energy per unit, packaging, line changeovers, fragmented deliveries and under-used kitchen capacity. These costs are not all equally marginal. The report therefore separates a technical incremental cost, a planning-level variable cost and a fully absorbed unit cost.

MAIN MODEL RESULT

Across the four recipes, average modelled variable-cost improvement is 49.3p per portion. At 10,000 portions a day for 260 operating days, that is about £1.28m a year before tax, financing, implementation costs or any pass-through adjustment. Every productivity improvement is an explicit pilot target, not an observed sector average.

What the evidence says

- Food-price pressure has moderated but not disappeared: ONS-linked data put annual food and non-alcoholic beverage inflation at 1.3% in September 2026. Individual category prices and supplier invoices remain heterogeneous. [1][2]
- Defra estimates the agri-food sector contributed £162.3bn of GVA in 2024 and employed 4.1m people in 2025. Food-chain productivity beyond the farmgate showed no significant improvement in 2024; catering productivity fell 1.7% and manufacturing fell 0.5%. [3]
- The case for process redesign is reinforced by food-waste estimates: WRAP reports 1.1m tonnes of hospitality and food-service waste each year, with a large share considered potentially edible. These are historical system estimates, not a claim that all waste can be recovered. [12]
- The food supply chain is not a pool of excess profit: an Oxford Economics/IGD study estimated typical net margins of 1-2% and 29p profit across a £20.24 basket of nine staple foods in 2025. Policy should target productivity and waste rather than assume price cuts can be funded by squeezing margins. [10]

Recommendations to HM Treasury

- Launch a 24-month, three-region Prepared Food Productivity Pilot with matched control kitchens and independently audited cost, quality and nutrition outcomes.
- Offer time-limited capital support or loan guarantees for modular cooking, portioning, sealing, rapid-chilling, energy monitoring and traceability equipment where suppliers commit to verified consumer-price pass-through.
- Use predictable public procurement demand from hospitals, schools, universities and other public kitchens to improve utilisation and lower ingredient and distribution costs, while preserving competitive tendering and SME access.
- Require open-book reporting of ingredient yields, labour minutes, energy per meal, waste, packaging, wholesale price and retail price so that any productivity gain can be shown to reach consumers.

Decision requested

Approve scoping and design of a pilot, not a blanket food subsidy. The proposed first-stage task is to validate supplier invoices, site energy, labour productivity, food-safety controls and retail economics with participating operators before committing scale-up capital.

1. UK prepared-food economics: the case for productivity

1.1 The market backdrop

The latest ONS release available at the report date covers September 2026. Food-price inflation has eased materially from the 2022-23 shock; ONS-linked September price data show annual food and non-alcoholic beverage inflation of 1.3%. This is a rate of change, not a measure of affordability: the price level remains higher than before the inflation surge, and consumers still face wide variation across foods and retailers. [1][2]

The food chain is economically material. Defra records £162.3bn of agri-food gross value added in 2024, including £39.3bn in food and drink manufacturing, £44.1bn in food retail and £46.1bn in non-residential catering. It also records 4.1m agri-food employees in 2025. [3]

1.2 Productivity is the lever

Defra reports no significant change in food-chain total factor productivity beyond the farmgate in 2024, compared with 0.4% growth in the wider economy. Manufacturing productivity fell 0.5%; catering fell 1.7%. These are sector-level statistics, not evidence that every kitchen is inefficient, but they indicate scope to test better systems for standardised preparation and throughput. [3]

1.3 Ingredient prices are only one layer

The price of a packaged meal includes farmgate or wholesale ingredients, inbound handling, preparation labour, energy, cooking yield, packaging, food safety and quality control, chill storage, outbound logistics, wastage, fixed overhead and the commercial terms of the selling channel. Defra identifies farmgate prices, import prices, exchange rates, food-manufacturing labour and non-labour manufacturing costs among the main drivers of food prices. [3]

1.4 Food waste and margin realities

WRAP's food-waste estimates put hospitality and food-service waste at about 1.1m tonnes annually; a WRAP statement has described approximately three-quarters as potentially edible and an estimated £3.2bn business cost. The underlying waste data refer to earlier years and should be treated as a system baseline rather than a current audited total for every operator. [12]

Conversely, a 2026 Oxford Economics/IGD analysis estimated UK grocery supply-chain net margins typically at 1-2%; it calculated that farmers, processors and retailers together made 29p profit from a £20.24 basket of nine everyday foods in 2025. A credible affordability programme should lower resource use per meal and strengthen supply resilience, not presume that supermarket margins alone can absorb the difference. [10]

SUPPLY-CHAIN CONSTRAINT

In 2025, 58% of UK food consumption by unprocessed farmgate value was of UK origin; 23% came from the EU and 19% from the rest of the world. UK production-to-supply was estimated at 60% for all food and 72% for indigenous-type foods. A stronger domestic prepared-food sector should use UK seasonal produce where efficient, while retaining diversified import channels for ingredients such as pollock, oil and some spices. [3]

2. Current UK price benchmarks

The table below juxtaposes public wholesale listings used in the recipes with ONS-linked consumer-price averages. They serve different purposes. Wholesale prices help estimate a producer's recipe input; consumer averages show the market context facing households. They are not directly interchangeable: pack size, quality, location, delivery, brand, origin and date all matter.

2.1 Selected wholesale ingredient prices

Ingredient	Reference price	Source note
Chicken thigh, boneless/skinless	£3.90/kg	JJ Foodservice public list; checked 7 Sep 2026
Beef mince	£7.50/kg	JJ Foodservice public list; checked 17 Oct 2026
Carrots / onions	£0.75/kg each	JJ Foodservice public list; Oct 2026
Potatoes	£0.68/kg	JJ Foodservice public list; 25 Oct 2026
Frozen peas	£2.79/kg	JJ Foodservice public list; 25 Oct 2026
Button mushrooms	£2.52/kg	JJ Foodservice public product page; accessed 5 Nov 2026
MSC Alaskan pollock fillets	£6.39/kg	JJ Foodservice public product page; accessed 5 Nov 2026
Chopped tomatoes / tomato purée	£1.13/kg / £2.50/kg	JJ Foodservice public list; Oct 2026
Whole milk / double cream	£0.85/L / £3.79/L	JJ Foodservice public list; 25 Oct 2026
Rapeseed oil / butter	£1.90/L / £6.00/kg	JJ Foodservice public list; 25 Oct 2026
Plain flour / grated cheddar	£0.81/kg / £6.00/kg	JJ Foodservice public list; Oct 2026
UK organic pearl barley	£1.80/kg	20 kg wholesale sack £36; Survival Wholefoods

The ChefSums compilation explicitly warns that its list is one supplier's public listing on a particular date, not a market average; it says actual invoices, delivery and account terms determine the buyer's price. We use it as a transparent costing proxy, not as a procurement guarantee. [4] Mushrooms and pollock are checked directly against JJ Foodservice pages. [5] Pearl barley is listed as UK-origin organic product by Survival Wholefoods. [6]

2.2 Consumer ready-meal price comparator

ONS-linked category average	Recorded pack size	Average shelf price	Implied price/kg
Meat ready meal	400 g	£3.43	£8.58/kg
Fish ready meal	400 g	£3.99	£9.98/kg
Vegetable ready meal	350 g	£2.18	£6.23/kg

These are experimental average prices from the ONS Shopping Prices comparison tool for September 2026, averaged across shops, brands and pack sizes by the underlying category. They are useful benchmarks, not a like-for-like quality comparison. [2]

2.3 Labour, energy and packaging inputs

Cost input	Reference	Model treatment
Minimum wage reference	£12.71/hour	National Living Wage for workers aged 21+ from 1 April 2026; model uses £15/hour fully loaded.
Electricity benchmark	27.67p/kWh	Q2 2026 small-to-medium business band (500,000-1,999,999 kWh/year), incl. CCL, excl. VAT; official DESNZ series.
Meal pack and lid	about £0.33	750 ml microwaveable PP round bowl and lid; 4+ case prices £56.15 + £43.09 for 300 sets, ex VAT.

The £15/hour labour rate is a modelled fully loaded cost above the statutory floor, allowing for employer on-costs and paid non-productive time; it is not a quoted wage. The 0.22 kWh per meal energy intensity is a pilot target. The energy price is a public Q2 2026 benchmark derived from DESNZ's quarterly data; site-specific contracts will differ. [7][8][9]

3. Cost framework and operating assumptions

3.1 Three meanings of marginal cost

In economics, marginal cost is the additional cost of producing one more unit. In a food factory, the answer depends on the time horizon and whether a new shift, worker, machine or delivery run is required. If the next tray uses spare capacity on a staffed line, the short-run marginal cost may be close to ingredients, packaging and incremental energy. If producing it requires another paid hour or another delivery, the marginal cost rises in steps.

Measure	Definition in this report
Recipe ingredient cost	Raw ingredient bill divided by 10 portions; then add 5% for trim, cooking loss and normal process waste.
Technical incremental cost	Recipe ingredients after the 5% allowance + meal tray/lid + process electricity.
Variable factory-delivered cost	Technical incremental cost + direct labour allocation + hygiene consumables + distribution/cold-chain allowance.
Fully absorbed unit cost	Variable factory-delivered cost + £0.30 illustrative fixed-site/capital allocation per portion.

IMPORTANT INTERPRETATION

The reported £1.30-£2.00 technical incremental costs are the closest approximation to adding one meal when labour and distribution capacity already exist. The £1.74-£2.45 variable factory-delivered figures are a planning model that spreads target labour productivity and normal logistics across each portion. The fully absorbed figures add a fixed-cost allowance; they are not pure marginal costs.

3.2 Operating assumptions

Input	Assumption
Recipe batch size	10 portions per recipe; intended finished portion about 450 g.
Food yield allowance	5% of raw ingredient cost in the high-throughput case; compared with 10% in the illustrative baseline.
Labour	£15 per paid labour-hour fully loaded; target 60 finished meals per paid labour-hour = £0.25/meal.
Energy	0.22 kWh per meal × £0.2767/kWh = about £0.06 per meal.
Packaging	£0.33 per meal for microwaveable, freezer-safe 750 ml PP bowl and lid, ex VAT.
Sanitation / consumables	£0.04 per meal, an explicit model allowance.
Cold-chain / outbound distribution	£0.15 per meal, an explicit consolidated-route allowance.
Fixed site / capital allocation	£0.30 per meal, illustrative rent, maintenance, supervision, depreciation and finance allowance.

3.3 What makes the model “Anglo-futurist”?

The menu is deliberately recognisable: chicken and barley, fish pie, shepherd’s pie and a mushroom-root pot. The futurism is in the production system rather than novelty ingredients: digital recipe control; precise check-weighing; whole-batch cooking; consistent portioning; traceable ingredients; rapid chilling; reusable data on yields and consumer demand; and a clean, high-quality presentation. The recipes favour British staples where price and quality support them, while avoiding a false claim that every input can be domestically sourced.

Recipe 1 | Yorkshire Orchard Chicken & Barley

British poultry, UK-grown barley, roots and peas; lemon and herbs lift the flavour without a costly sauce. Target finished portion: approximately 450 g.

Ingredient	Batch quantity	Cost basis	Batch cost
Boneless, skinless chicken thigh	1.60 kg	£3.90/kg	£6.24
UK organic pearl barley	0.80 kg	£1.80/kg	£1.44
Carrots	0.70 kg	£0.75/kg	£0.53
Onions	0.40 kg	£0.75/kg	£0.30
Frozen peas	0.50 kg	£2.79/kg	£1.40
Garlic	0.05 kg	£3.49/kg	£0.17
Rapeseed oil	0.10 L	£1.90/L	£0.19
Lemon, herbs, stock & seasoning allowance	batch allowance	model allowance	£0.99
TOTAL RAW INGREDIENTS	10 portions		£11.25

Cost measure	Per 450 g portion
Raw ingredients per portion	£1.13
Ingredient cost incl. 5% trim / waste	£1.18
Technical marginal cost (ingredients + pack + energy)	£1.57
Variable factory-delivered cost	£2.01
Fully absorbed cost (incl. £0.30 fixed allocation)	£2.31

Method

Sweat the onion and garlic in rapeseed oil. Toast the barley briefly, then simmer in light stock until tender but still structured. Roast or gently braise the chicken thigh with lemon and sage; roast the carrots separately to retain sweetness. Fold peas through the barley near the end. Portion the barley, vegetables and chicken evenly into 10 trays. Chill rapidly, seal and label.

Product and production notes

- UK barley provides a domestic grain base; poultry origin and assurance standard should be specified in the procurement contract.
- Contains barley (gluten); no intentional dairy is specified, but shared-line cross-contact and stock formulation must be checked.
- Recipe quantities are pilot formulations, not validated commercial specifications. A process technologist and registered dietitian should confirm final yield, nutritional profile, allergen declaration and shelf life.

Recipe 2 | Atlantic Pollock & Garden Pea Fish Pie

A generous fish-pie format using MSC-certified frozen pollock, a traditional potato topping and a light cream sauce. Target finished portion: approximately 450 g.

Ingredient	Batch quantity	Cost basis	Batch cost
MSC Alaskan pollock, skinless and boneless	1.40 kg	£6.39/kg	£8.95
White potatoes	2.30 kg	£0.68/kg	£1.56
Frozen peas	0.70 kg	£2.79/kg	£1.95
Double cream	0.30 L	£3.79/L	£1.14
Whole milk	0.40 L	£0.85/L	£0.34
Butter	0.12 kg	£6.00/kg	£0.72
Plain flour	0.05 kg	£0.81/kg	£0.04
Onions	0.30 kg	£0.75/kg	£0.23
Rapeseed oil	0.05 L	£1.90/L	£0.10
Herbs, lemon, stock & seasoning allowance	batch allowance	model allowance	£0.35
TOTAL RAW INGREDIENTS	10 portions		£15.37

Cost measure	Per 450 g portion
Raw ingredients per portion	£1.54
Ingredient cost incl. 5% trim / waste	£1.61
Technical marginal cost (ingredients + pack + energy)	£2.01
Variable factory-delivered cost	£2.45
Fully absorbed cost (incl. £0.30 fixed allocation)	£2.75

Method

Boil the potatoes and mash with part of the butter and milk. Sweat diced onion, make a light roux with flour and butter, then loosen with milk and cream. Poach the pollock gently in the sauce; fold through peas and herbs. Fill 10 trays and top with mash. Bake until the centre is fully cooked, then rapidly chill, seal and label. Validate the fish allergen controls and final cooking/chilling regime in the production HACCP plan.

Product and production notes

- Use the frozen MSC pollock case price as a wholesale proxy; secure a delivered quote and verify fish species, MSC chain-of-custody, country of origin and glazing percentage before contract.
- Contains fish, milk and wheat (flour); control allergen segregation, label accuracy and supplier substitution.
- Recipe quantities are pilot formulations, not validated commercial specifications. A process technologist and registered dietitian should confirm final yield, nutritional profile, allergen declaration and shelf life.

Recipe 3 | Yorkshire Beef & Mushroom Shepherd's Pie

A premium but recognisably British cottage-pie profile, with mushrooms and barley adding depth and helping extend the meat component without turning the meal into a low-meat substitute. Target finished portion: approximately 450 g.

Ingredient	Batch quantity	Cost basis	Batch cost
Beef mince	1.00 kg	£7.50/kg	£7.50
Button mushrooms	0.60 kg	£2.52/kg	£1.51
White potatoes	2.00 kg	£0.68/kg	£1.36
Carrots	0.50 kg	£0.75/kg	£0.38
Onions	0.50 kg	£0.75/kg	£0.38
UK organic pearl barley	0.15 kg	£1.80/kg	£0.27
Chopped tomatoes	0.50 kg	£1.13/kg	£0.56
Tomato purée	0.08 kg	£2.50/kg	£0.20
Grated mature cheddar	0.15 kg	£6.00/kg	£0.90
Whole milk	0.20 L	£0.85/L	£0.17
Butter	0.10 kg	£6.00/kg	£0.60
Rapeseed oil	0.05 L	£1.90/L	£0.10
Plain flour	0.04 kg	£0.81/kg	£0.03
Herbs, stock & seasoning allowance	batch allowance	model allowance	£0.30
TOTAL RAW INGREDIENTS	10 portions		£14.25

Cost measure	Per 450 g portion
Raw ingredients per portion	£1.43
Ingredient cost incl. 5% trim / waste	£1.50
Technical marginal cost (ingredients + pack + energy)	£1.89
Variable factory-delivered cost	£2.33
Fully absorbed cost (incl. £0.30 fixed allocation)	£2.63

Method

Brown the beef in batches to avoid stewing. Sweat onions, carrots and mushrooms; add barley, tomatoes, purée, stock and herbs, then simmer until thick and the barley is tender. Boil and mash the potatoes with milk and butter. Portion the filling into 10 trays, top with mash and a measured cheddar finish, then bake. Rapidly chill, seal and label.

Product and production notes

- Beef mince price is a single supplier public-listing benchmark; confirm fat percentage, origin, welfare specification and delivered yield on invoice.
- Contains barley (gluten), wheat and milk; the production line must prevent cross-contact with fish and other allergens.
- Recipe quantities are pilot formulations, not validated commercial specifications. A process technologist and registered dietitian should confirm final yield, nutritional profile, allergen declaration and shelf life.

Recipe 4 | Lincolnshire Barley, Mushroom & Root Pot (vegan)

A substantial plant-based meal built around UK-grown barley, mushrooms, peas and root vegetables. It is designed to be a main course, not a small side-dish portion. Target finished portion: approximately 450 g.

Ingredient	Batch quantity	Cost basis	Batch cost
UK organic pearl barley	0.80 kg	£1.80/kg	£1.44
Button mushrooms	1.00 kg	£2.52/kg	£2.52
Carrots	0.70 kg	£0.75/kg	£0.53
Onions	0.50 kg	£0.75/kg	£0.38
White potatoes	1.20 kg	£0.68/kg	£0.82
Chopped tomatoes	0.80 kg	£1.13/kg	£0.90
Frozen peas	0.50 kg	£2.79/kg	£1.40
Garlic	0.05 kg	£3.49/kg	£0.17
Rapeseed oil	0.10 L	£1.90/L	£0.19
Herbs, stock & seasoning allowance	batch allowance	model allowance	£0.30
TOTAL RAW INGREDIENTS	10 portions		£8.64

Cost measure	Per 450 g portion
Raw ingredients per portion	£0.86
Ingredient cost incl. 5% trim / waste	£0.91
Technical marginal cost (ingredients + pack + energy)	£1.30
Variable factory-delivered cost	£1.74
Fully absorbed cost (incl. £0.30 fixed allocation)	£2.04

Method

Cook the barley in a steam kettle or pressure-assisted batch cooker. Sear mushrooms in a little rapeseed oil to build colour; add onions, carrots, potatoes, garlic and tomatoes with stock and herbs. Simmer until the vegetables are tender and the sauce is cohesive. Fold through peas and barley. Finish with herb oil, then portion, rapidly chill, seal and label.

Product and production notes

- Vegan claim requires ingredient and process verification, including stock, flavourings, oils and shared-line cross-contact controls.
- Contains barley (gluten). This dish provides a lower ingredient-cost anchor while retaining a large, satisfying portion.
- Recipe quantities are pilot formulations, not validated commercial specifications. A process technologist and registered dietitian should confirm final yield, nutritional profile, allergen declaration and shelf life.

5. Unit economics and the marginal-cost opportunity

The recipe-level figures below use the same operating assumptions across all four meals. This allows a fair comparison of food composition and the effect of a standardised production process. Amounts are rounded to the nearest penny; actual invoices and measured yields should replace every model input before investment.

Meal	Ingredients after 5%	Technical incremental	Variable delivered	Fully absorbed	Modelled saving
Chicken & barley	£1.18	£1.57	£2.01	£2.31	£0.49
Pollock fish pie	£1.61	£2.01	£2.45	£2.75	£0.51
Beef & mushroom pie	£1.50	£1.89	£2.33	£2.63	£0.50
Vegan barley pot	£0.91	£1.30	£1.74	£2.04	£0.47

Technical incremental = ingredients + package + energy. Variable delivered adds labour, hygiene and distribution. Fully absorbed adds the £0.30 fixed-cost allocation. Savings compare the illustrative baseline with the high-throughput variable delivered model, not with a national industry average.

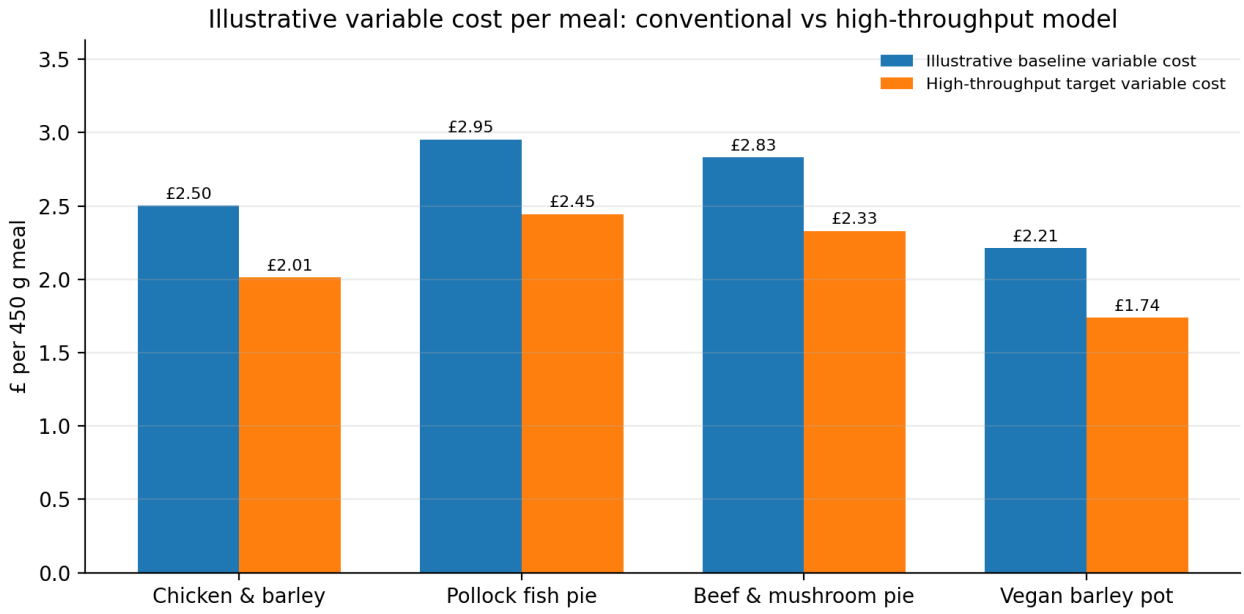


Figure 1. Illustrative per-meal variable cost under the stated baseline and high-throughput assumptions. The baseline is a constructed comparator (30 meals per paid labour-hour, 10% waste, 0.44 kWh per meal, 25p distribution and 6p hygiene); it is not a statistical estimate of the average UK kitchen.

5.1 Why the model saves about 49p

Lever	Average saving per meal
Labour productivity: 30 to 60 meals per paid hour	25p
Energy intensity: 0.44 to 0.22 kWh per meal	6p
Consolidated distribution: 25p to 15p per meal	10p
Sanitation / consumables allowance: 6p to 4p	2p
Lower ingredient loss: 10% to 5%	about 6p
Total indicative improvement	about 49p

The largest single lever is paid labour productivity. This is a stretch target, not a forecast: the result must be measured from rosters, line-hours, changeovers, downtime, quality rejects and actual sellable meals. Better automation should remove repetitive movement and improve consistency, not simply intensify work or displace skilled food staff without a transition plan.

5.2 Scale arithmetic

Operating scale	Meals/year (260 days)	Saving at modelled 49p
1,000 meals/day	260,000	£128,121
5,000 meals/day	1,300,000	£640,604
10,000 meals/day	2,600,000	£1,281,209
100 million meals/year (illustrative scale arithmetic)	100,000,000	£49,277,263

The 100m-meal figure is not a demand forecast. It shows only the arithmetic if the same saving were independently validated across that volume. Implementation costs, finance, taxes, extra quality assurance, distribution distance and any rebound in ingredient prices would reduce the realised benefit.

5.3 Sensitivity

Sensitivity test	Approximate effect
Ingredient prices +10%	Adds about 9p-16p per portion, depending on recipe.
Labour productivity only 30 meals/hour	Adds 25p per portion against the target case.
Energy use doubles to 0.44 kWh	Adds about 6p per portion at the chosen tariff.
Food loss rises from 5% to 10%	Adds around 4p-8p per portion across the recipes.
No distribution consolidation	A 10p-per-meal increase removes roughly one-fifth of the expected saving.

6. Consumer price proposition and channel economics

The aim is not merely to make an inexpensive meal; it is to offer an attractive, full-size, reliable meal at a price that is competitive with today's prepared-food aisle. The proposed price points below are hypotheses for a live pilot. Product quality, recipe size, taste, nutrition, packaging and brand positioning must be tested with consumers.

Meal	Pack	Test shelf price	Price/kg	ONS-linked comparison
Chicken & barley	450 g	£3.49	£7.76/kg	Meat ready meal avg: £8.58/kg
Pollock fish pie	450 g	£4.29	£9.53/kg	Fish ready meal avg: £9.98/kg
Beef & mushroom pie	450 g	£3.79	£8.42/kg	Meat ready meal avg: £8.58/kg
Vegan barley pot	450 g	£2.99	£6.64/kg	Vegetable ready meal avg: £6.23/kg

The meat, fish and vegetable comparators are September 2026 average prices for different pack sizes and broad categories, not direct quality equivalents. The proposed meals are modelled at a uniform 450 g. The plant-based pot is intentionally priced as a premium full meal rather than trying to undercut every vegetable ready meal. [2]

6.1 Illustrative manufacturer economics

For a simple channel illustration, assume a retailer retains 25% of the shelf price before the producer receives the remaining 75%. This is a modelling convention, not an observed universal retailer margin or a negotiated wholesale term. It excludes promotions, rebates, listing fees, returns, distribution agreements and payment terms.

Meal	Shelf price	Producer net at 75%	Fully absorbed cost	Contribution before central costs
Chicken	£3.49	£2.62	£2.31	£0.30
Fish	£4.29	£3.22	£2.75	£0.47
Beef	£3.79	£2.84	£2.63	£0.21
Vegan	£2.99	£2.24	£2.04	£0.20

The contribution shown is not net profit. It is the difference between an assumed producer receipt and the modelled fully absorbed unit cost before head-office overhead, marketing, financing, tax, product development, retailer promotions and any non-standard logistics. The beef and vegan lines are relatively tight under this assumption; they may need a higher realised wholesale price, direct institutional sales, a lower cost contract or a revised recipe after the pilot.

6.2 VAT treatment

The report's shelf-price illustration assumes a chilled, packaged meal sold as a food product for reheating at home and eligible for zero-rating. HMRC states that most food of a kind used for human consumption is zero-rated, while food supplied in the course of catering, including hot takeaway food, is standard-rated. The precise treatment depends on product presentation and supply circumstances; obtain tax advice before launch. [11]

7. The regional production model

A “Mealworks” hub would operate as a small number of regional, high-utilisation food factories rather than a large number of lightly loaded kitchens. A hub should be sized to a proven buyer base and designed around modular cells, so a region can expand capacity without building a single monolithic plant.

7.1 Process architecture

Stage	Operating design
1. Forecast demand	Aggregate retailer, institutional and subscription orders by SKU and delivery day; use a conservative forecast and a minimum committed order book.
2. Procure and receive	Pool orders for meat, fish, vegetables, grains and packaging; check supplier lot, origin, temperature, weight and invoice price.
3. Prepare ingredients	Mechanised washing, peeling and dicing where appropriate; weigh recipe inputs; collect trim data and segregate safe by-products.
4. Cook in controlled batches	Steam kettles, combi ovens and pressure-assisted cookers; digital time/temperature control; standardised recipes and batch logs.
5. Portion, seal and verify	Automated or semi-automated filling; check-weigh every tray; seal; metal-detect or apply other risk-based controls; code each lot.
6. Rapid chill and dispatch	Validated blast-chill or equivalent process; temperature logging; short regional routes; FEFO inventory and traceability.
7. Learn and re-cost	Feed actual yields, rejects, energy, labour, sell-through and waste back into recipe costing and next-day production planning.

7.2 Design principles

- Standardise a small core of ingredients across recipes so purchasing, stock control and trim utilisation improve without making the menu monotonous.
- Use seasonal UK produce when the delivered cost, quality and yield are competitive; retain diversified import sources for foods not reliably produced domestically.
- Build cold-chain and food-safety controls into the design from the start: validated cook/chill parameters, allergen segregation, temperature records, shelf-life testing and batch traceability.
- Design for labour augmentation: automated lifting, weighing, dicing and tray handling should reduce repetitive tasks while retaining skilled staff for quality, maintenance, menu development and food-safety oversight.
- Use transparent unit-level measurement. A hub should know the measured cost of each sellable meal, not just the average cost of a batch.

7.3 Demand anchors

Public procurement can create the regular baseline volume that makes high utilisation possible. Potential buyers include NHS and care catering, schools and colleges, local authority meal services, universities and large employers. Procurement should be competitive, quality-led and open to regional SMEs supplying produce, meat, dairy and prepared components. It should not guarantee a single national supplier or lock local producers out.

8. Policy programme for HM Treasury

8.1 Proposed 24-month pilot

Phase	Delivery
Months 0-3: design and baseline	Select three candidate regions; map public and private demand; collect anonymised current kitchen costs; specify nutrition, quality, allergen and carbon measures; agree independent evaluation.
Months 4-9: procurement and fit-out	Competitive selection of operators; supplier framework agreements; lease or retrofit modular cooking, filling, sealing and rapid-chilling equipment; train staff; validate HACCP and shelf-life protocols.
Months 10-18: live operation	Run at measured throughput; compare with matched conventional kitchens; publish quarterly price, labour, energy, yield, waste and consumer acceptance metrics.
Months 19-24: independent decision	Audit whether unit-cost savings persisted, whether they reached consumers, whether nutrition and quality standards were met, and whether the model can be expanded without concentration risk.

8.2 Treasury instruments

Instrument	Design guardrail
Capital allowances / investment support	Time-limited support for equipment that demonstrably reduces energy, waste or labour minutes per saleable meal. Require baseline measurement and clawback for non-delivery.
Loan guarantees	Share early equipment risk for viable regional hubs with contracted demand; avoid underwriting speculative capacity without buyers.
Public procurement framework	Enable multi-year, competitively tendered demand lots; make whole-life cost, nutrition, taste, traceability and resilience part of the award.
Open measurement standard	Create a common cost-per-meal reporting template: ingredient yield, packaging, kWh, paid labour minutes, waste, cold-chain, supplier payments and retail pass-through.
Competition safeguards	Use multiple regional suppliers, open-book costing, data portability, non-exclusive contracts and regular retendering to prevent local monopsony or a national duopoly.

8.3 Pilot scorecard

Measure	Proposed threshold / reporting
Variable delivered cost	At least 40p/meal below matched baseline after stabilisation; 49p is the central scenario, not an automatic pass mark.
Paid labour productivity	Target 60 saleable meals per paid production labour-hour, adjusted transparently for recipe complexity and maintenance downtime.
Ingredient loss	Target at or below 5% of ingredient cost/weight using a consistent audited definition.
Energy	Target at or below 0.22 kWh per saleable 450 g meal, measured at site meter and reconciled to output.
Price pass-through	Publish changes in producer net price and retail shelf price; require agreed proportion of verified savings to reach buyer/consumer.
Quality and nutrition	Blind taste scores, portion weights, nutrition panel, allergen accuracy, complaint rate, microbiological compliance and shelf-life validation.
Workforce and suppliers	Track pay, training, job quality, SME supplier access, UK-origin share and payment terms.

9. Risks, constraints and mitigations

Risk	Why it matters	Mitigation
Food safety and chilled shelf life	A larger plant concentrates risk if cook/chill, temperature or allergen controls fail.	Validate every recipe and pack format; independent microbiological shelf-life tests; HACCP; traceability; recall drills; temperature telemetry.
Commodity and supplier volatility	Beef, poultry, fish, dairy, vegetables, energy and packaging prices may move faster than a fixed menu price.	Dual-source critical ingredients; use index-linked contracts with transparent reset rules; publish recipe cost variance.
Overstated labour productivity	60 meals per paid hour may be unattainable after cleaning, changeovers, maintenance, training and downtime.	Measure net saleable throughput over full shifts; include cleaning and changeover time; scale only after measured performance.
Retailer bargaining power	A concentrated buyer may capture the saving rather than reduce shelf prices.	Multiple channels; open-book terms; institutional procurement; publish price pass-through; preserve alternative routes to market.
Nutritional drift	Cheap formulation can increase salt, saturated fat or refined carbohydrate, or shrink protein portions.	Set minimum protein and fibre ranges; monitor salt and saturated fat; use dietitian sign-off and blind taste panels.
Waste transfer rather than waste prevention	A factory could reduce kitchen scraps but increase unsold finished stock.	Forecast demand; FEFO stock; small-batch flexibility; donate safe surplus; measure total system waste from farmgate input to consumer discard.
Packaging and cold-chain burden	Lower-cost plastic packaging or extra chilled distribution may undermine environmental goals.	Test recyclable/reusable formats where safe; measure total packaging weight and life-cycle impacts; optimise route density.
Market concentration and SME exclusion	Scale economies may favour large manufacturers and displace local food businesses.	Regional lots; modular capacity; cooperative access; open contracts; subcontracting opportunities; publish supplier concentration metrics.

9.1 What would falsify the proposition?

The proposal should be stopped or redesigned if independent measurement shows that the model cannot maintain taste, nutritional quality, food safety and reliable delivery while reducing cost per sellable meal; if the cost saving disappears once full shift coverage and cleaning are included; or if wholesale cost reductions do not reach consumers or public buyers. A pilot is valuable precisely because it can reject the model before public money is committed at scale.

10. Conclusion and decision

The United Kingdom does not need a low-quality prepared-food economy to achieve a lower price per meal. It needs a more productive one. The four recipes in this report demonstrate that familiar, good-quality meals can be designed around a relatively economical core of poultry, barley, roots, peas, mushrooms and potatoes, while retaining attractive fish and beef options. Their illustrative ingredient costs range from around 91p to £1.61 per 450 g portion after a 5% loss allowance.

The bigger opportunity lies in the system surrounding the recipe: buying at scale, reducing trim and spoilage, moving from small fragmented batches to well-utilised production cells, using precise portion control, reducing energy per meal, and delivering dense regional routes. Under explicit assumptions, the cost model produces approximately 49p of variable-cost headroom per meal. At 10,000 portions a day across 260 days, that equates to around £1.28m per year at one hub.

That arithmetic is promising but conditional. It is not evidence that the average UK meal can immediately be produced at these costs, nor that a 49p saving can be passed fully through to the till. Actual results depend on supplier invoices, yields, recipe validation, equipment utilisation, staffing, finance, selling terms and the distance between production and customer.

DECISION FOR HM TREASURY

Commission a 24-month, three-region Prepared Food Productivity Pilot with independent baseline measurement, open-book costing and a binding consumer/public-buyer pass-through test. Back proven capital productivity, not an untargeted food subsidy. Scale only if the pilot demonstrates lower cost per saleable meal alongside safe food, strong nutrition, high taste scores, fair workforce outcomes and resilient supplier competition.

Appendix A. Costing method and calculation checks

All recipe amounts represent one 10-portion production batch. For each dish, the raw ingredient bill is divided by 10. The high-throughput ingredient cost per portion is then multiplied by 1.05 to allow 5% for trim, cooking loss and normal process waste. The comparison baseline uses a 10% allowance. The same 5% and 10% assumptions are applied across the four dishes solely to enable a controlled illustrative comparison.

Model line	Formula
Packaging	(£56.15 + £43.09) ÷ 300 = £0.3308 per meal, rounded to £0.33.
Energy	0.22 kWh × £0.2767/kWh = £0.0609, rounded to £0.06.
Labour	£15 fully loaded per paid hour ÷ 60 saleable meals = £0.25 per meal.
Hygiene / consumables	£0.04 per meal (assumption).
Distribution / cold-chain	£0.15 per meal (assumption).
Fixed allocation	£0.30 per meal (assumption, excluded from short-run marginal cost).
Target variable cost	Ingredient cost after 5% allowance + packaging + energy + labour + hygiene + distribution.
Baseline variable cost	Raw ingredients × 1.10 + packaging + 0.44 kWh × £0.2767 + £0.50 labour + £0.06 hygiene + £0.25 distribution.

All prices are expressed in pounds sterling. Supplier prices may exclude delivery and VAT. Food ingredients generally have distinct VAT treatment from packaging, energy and services; the costing table is an ex-VAT producer cost model. The recipe costing assumes an appropriate zero-rating position for most basic food inputs, but input VAT recovery and mixed supplies must be reviewed by the operator's accountant.

Appendix B. Data limitations

- Public wholesale prices are snapshots from a small number of supplier pages, not a statistically representative UK wholesale basket. Contract rebates, account pricing, delivery charges, pack specifications and regional availability can change the result.
- The ONS-linked ready-meal prices are averages across brands, retailers and sizes. The broad category products are not matched for recipe, quality, weight, nutrition or packaging.
- The labour productivity, waste, energy intensity, sanitation, distribution and fixed-cost values are scenario assumptions or targets unless linked above to a source. They are not claimed as current observed averages.
- The report estimates production economics, not a full business plan. It does not include capex, depreciation schedules beyond a simple 30p allocation, financing, central overhead, product development, tax, insurance, retailer fees, promotional funding or working-capital costs.
- No nutritional analysis or commercial food-safety validation has been performed. The recipes are costed concepts requiring professional testing before sale.

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